Closing the Physical-to-Chain Gap: Tamper-Evident Provenance Attestation for Additive Manufacturing Supply Chains

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***Abstract*—Blockchain provenance for additive manufacturing (AM) is only as trustworthy as the bridge that converts physical process data into on-chain commitments, an instance of the oracle problem that existing AM provenance work leaves largely unaddressed. We present a threat-model-driven attestation architecture for the printer-to-ledger path: a four-tier trust ladder (T0 naïve submission, T1 edge-signed telemetry bundles, T2 on-chain device registry, T3 hardware root of trust) and a canonical device-signed bundle binding the certified design (G-code SHA-256), commanded process envelope, observed telemetry, and job outcome into one Ed25519-signed unit whose hash is committed on-chain. We implement the T1 tier end to end on a Creality Ender-3 with OctoPrint and Hyperledger Fabric v3.1.0, coupling measured physical adherence to an on-chain Bayesian reputation system within a single atomic transaction. Across five real print jobs spanning three geometries and three outcome types, a controlled $+15\,^{\circ}$C setpoint injection lowered the derived quality score by 9.2% while design-hash compliance held at 1.0, so the bridge separates faithful execution from design substitution from telemetry alone. Each job produced a 7–14 kB signed bundle anchored by a constant 369 B on-chain footprint, the attestation pipeline runs in 1.55 ms, and five adversarial tests against the T1 defenses, including a live design-substitution print, were each detected or rejected. Edge signing makes the evidence tamper-evident but does not prevent fabrication at source by a colluding device-key holder, the gap the T2 and T3 tiers address.**

***Index Terms*—additive manufacturing, blockchain, provenance, oracle problem, attestation, Hyperledger Fabric, reputation systems, supply chain security**

I. Introduction

Additive manufacturing (AM) supply chains increasingly rely on permissioned blockchains for tamper-evident lifecycle records spanning material certification, printing, inspection, and final certification [1], [2]. Our prior work anchors such records with low overhead [3] and couples them to adversarially robust on-chain reputation scoring [4]. These systems rest on a rarely examined assumption: that the data arriving at the chaincode interface faithfully describes what happened on the factory floor. A provenance ledger fed by an unauthenticated HTTP POST is an instance of the *oracle problem* [5]. It guarantees integrity after commitment but says nothing about the process that produced the data. In AM, sabotage attacks that alter process parameters produce parts that pass visual inspection yet fail in service [6], [7].

This paper studies the *physical-to-chain bridge*, the software and hardware path from the printer's sensors to the ledger commitment. We make four contributions. First, a four-tier trust analysis (Section III) covering T0 (naïve), T1 (edge-signed bundles), T2 (on-chain device registry), and T3 (hardware root of trust). Second, a canonical device-signed telemetry bundle (Section IV) that binds the certified design hash, commanded envelope, observed telemetry, and job outcome into one Ed25519-signed unit whose SHA-256 is the sole on-chain commitment. Third, an end-to-end T1 implementation (Section V) on a Creality Ender-3 with Hyperledger Fabric v3.1.0, evaluated on five real print jobs across three geometries and three outcome types, including a controlled thermal-anomaly injection. Fourth, an evaluation of physical adherence as a reputation signal (Section IV-C), mapped onto the $[0, 1]$ rating consumed by an on-chain Bayesian reputation engine within one atomic transaction.

II. Related Work

Blockchain provenance and reputation for AM. Traceability frameworks for AM record lifecycle events on permissioned ledgers, typically Hyperledger Fabric [8]. Alkhader et al. [1] propose smart contracts for AM part traceability, and Kurpjuweit et al. [2] survey blockchain adoption across AM supply chains. Our prior work anchors AM lifecycle events on Fabric with a low per-event footprint [3] and scores actor behavior from those events with an adversarially robust on-chain reputation layer [4]. These systems, ours included, accept submitted event data at face value. The chaincode validates the submitter's identity and the legality of the lifecycle transition, but not whether the reported physical process occurred. This paper addresses that residual gap, the integrity of the bridge feeding the ledger rather than the ledger itself.

The oracle problem, AM attacks, and trust signals. This gap is an instance of the blockchain oracle problem [5], in which external data carriers reintroduce the trusted third party that the ledger was meant to remove. Oracle networks such as Chainlink [9] aggregate independent reporters, but

manufacturing telemetry has exactly one witness, the machine itself, so the problem becomes authenticating that witness.

Town Crier [10] authenticates web sources using SGX [11] enclaves; our T3 tier (Section III-B) is the AM-floor analogue, while T1 and T2 are deployable intermediates that need no trusted hardware on legacy printers. Yampolskiy et al. [7] provide a taxonomy of sabotage through manipulated process parameters, and the dr0wned attack [6] produced a drone propeller that failed in flight after a design-file modification, which motivates our binding of the executed design hash and observed process envelope into one signed record. On-chain reputation systems typically score subjective ratings through Beta-distribution models [12], [13] or transitive peer trust [14], whereas our bridge derives the rating from measured physical adherence protected by the attestation layer. To our knowledge, no prior AM-blockchain work implements and evaluates a device-signed, canonically hashed telemetry attestation path on real printer hardware with an explicit trust boundary.

III. Threat Model and Trust Tiers

A. Setting and Actors

We consider a multi-organization AM supply chain in which a print bureau executes jobs against designs certified earlier in the lifecycle, and downstream consumers (inspectors, certifiers, buyers) rely on the on-chain record. The bridge path has four elements. The printer and its firmware report hotend and bed temperatures and job state over serial; consumer printers such as the Ender-3 have no secure element and no firmware attestation. The bridge host, an edge device (a Raspberry Pi), runs the print server, the telemetry recorder, and, at T1 and above, the device signing key. The print bureau is the Fabric member (MSP) operating the printer, motivated to appear reliable because its on-chain reputation gates future work [4]. The network path from bridge host to peer is TLS-protected gRPC in our deployment.

The Fabric layer (ordering, endorsement, validation) is assumed correct, and consensus attacks are out of scope. The adversary of interest is an insider or intruder on the bridge path who wants the ledger to show a conforming print that did not occur as recorded. Such an adversary has several avenues, each addressed by a tier in Table I: editing stored telemetry after submission (post-hoc tampering), replaying a good print's evidence against a different part, executing a non-certified G-code file (design substitution), feeding the recorder false samples or synthesizing a bundle outright (fabrication at source), reusing an exfiltrated signing key, or controlling host, key, and MSP identity at once (collusion).

B. Trust Tiers

We organize bridge designs into four cumulative tiers, each extending the guarantees of the previous one; Table I records what each tier does and does not guarantee.

1) The T1 trust boundary: T1 provides tamper-evidence after signing; it does not establish truth at the source. An adversary who controls the bridge host, and therefore the device key, can synthesize a plausible telemetry bundle, sign it, and commit its hash. Nothing in T1 or T2 distinguishes this from an honest capture, because the signature attests to the key rather than to the physical process. T1 remains the appropriate first rung for legacy fleets, since it converts the threats that dominate in practice, retroactive log editing, evidence reuse, and quiet design substitution, into cryptographically detectable events with no hardware changes. The fabrication-at-source residual is the gap that T3 targets (Section VII).

Two further properties come from the chaincode. The lifecycle state machine rejects a `PRINT_JOB` event for an asset not in the `MATERIAL_CERTIFIED` stage, so replayed evidence cannot attach to an arbitrary asset. And because the bundle's rating updates the operator's Beta posterior atomically with the provenance event (Section IV-C), committing only flattering evidence leaves a sparse, low-confidence history that the Wilson confidence gate penalizes.

IV. Bridge Architecture

Figure 1 shows the pipeline. Every field the verifier needs is contained in one signed unit, the bundle, or derivable from it, and a single 32-byte commitment crosses onto the ledger.

A. Canonical Signed Bundle

The bundle binds five blocks into one signed unit. The `design` block is the commanded process envelope derived from the certified G-code: its identity (SHA-256 over the raw bytes), the first commanded nozzle and bed temperatures (`M104`/`M109`, `M140`/`M190`), layer count, estimated duration, and filament length from the slicer header. The envelope records what the printer was instructed to do, so a verifier can detect divergence without the original file. The remaining blocks are `observed` (the telemetry samples, the executed G-code hash, and the actual duration), `assessment` (Section IV-B), `outcome` (the terminal job event), and the capture `window`. An `attestation` block carries the Ed25519 `signature`, the signer's `device_id`, and its raw public key.

Serialization is canonical JSON (sorted keys, no insignificant whitespace, UTF-8), so every verifier, including future T2 chaincode on a Fabric peer, recomputes byte-identical encodings. The device key signs $\mathrm{canon}(\texttt{payload})$, and the on-chain commitment (the event's `offChainDataHash`) is

$$h_{\mathrm{oc}} = H\big(\mathrm{canon}(\{\texttt{payload}, \texttt{attestation}\})\big), \qquad (1)$$

where H is SHA-256. Because the hash covers the signature itself, neither block can be substituted after submission, and verification reduces to two independent checks: recomputing Eq. (1) against the ledger value, and verifying the signature over the canonical payload (Section VI-D).

B. Observed-versus-Commanded Assessment

The assessment maps physical adherence onto three $[0, 1]$ rating dimensions. `quality` q is the fraction of nozzle and bed samples within a tolerance band ($\pm 8\,^{\circ}\mathrm{C}$ here) of the commanded envelope; `compliance` c is 1 if the executed G-code hash equals the certified design hash and 0 otherwise; and `delivery` is $d = \min\big(1, \max(0,\, 2 - t_{\mathrm{obs}}/t_{\mathrm{est}})\big)$, where

TABLE I
TRUST-TIER LADDER FOR THE PHYSICAL-TO-CHAIN BRIDGE. EACH TIER IS CUMULATIVE OVER THE PREVIOUS ONE.

Tier	Mechanism	Defends against	Does NOT defend against
T0	Naïve submission: client POSTs event fields under an org MSP identity	Nothing on the bridge path; only on-chain immutability after commit	Any manipulation before or at submission; evidence unauthenticated, unbound to the artifact
T1	Edge-signed canonical bundle: device Ed25519 key signs the payload; SHA-256 committed on-chain (**this paper**)	Post-hoc tampering (dual-detect); cross-asset replay (hash binding); design substitution; in-flight modification	Fabrication at source by a device-key holder; key theft from the unhardened host; bureau/host collusion
T2	On-chain device registry: chaincode verifies the bundle signature at endorsement; key enrollment and revocation are ledger events	Unregistered or revoked signing keys; bounds a stolen key's validity window; auditable device identity	Fabrication at source by the *registered* key holder; registry authenticates the key, not the physics
T3	Hardware root of trust: key in TEE or secure element [11], measured boot, sensor-level or threshold attestation. Strongest form needs printer-side hardware the consumer fleet lacks	Compromised host OS; key extraction; recorder substitution; raises fabrication to hardware-attack cost	Physical-world spoofing by a fully malicious organization (e.g. detached thermistors)

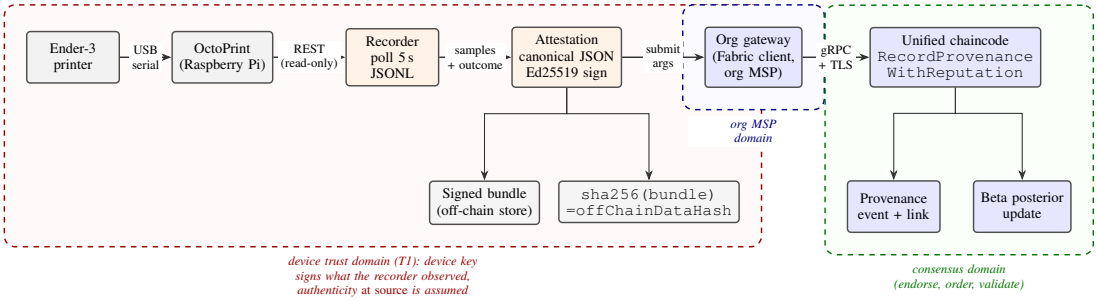


Fig. 1. The physical-to-chain bridge: a read-only recorder polls OctoPrint, the attestation module signs the canonical bundle and commits its SHA-256 (`offChainDataHash`), and one atomic chaincode call records the event and updates the operator's Beta posterior. Dashed boxes are trust domains; T1 is the device domain.

t_{obs} is the capture-window duration (warm-up included) and t_{est} the slicer estimate, independent of outcome. The scalar rating submitted with the event is

$$r \;=\; 0.6\,q + 0.25\,c + 0.15\,d. \tag{2}$$

Thermal adherence serves here as a concrete instance of a physical signal propagating verifiably onto the ledger; the security claim concerns faithful capture and tamper-evident commitment of that signal rather than the part's fitness for use. The weights are policy parameters ordered by security relevance: `quality` is the only continuous signal and therefore the dimension a fabrication or drift attack must move; `compliance` is binary and vetoes the rating on a design mismatch; `delivery` is the least security-relevant.

C. On-Chain Integration

The ledger layer is a three-contract Fabric chaincode (provenance, reputation, and integration) that the bridge drives as a client. A single `RecordProvenanceWithReputation` invocation runs the stake-gated reputation gate, appends the lifecycle event carrying `offChainDataHash` (advancing the asset `MATERIAL_CERTIFIED` → `PRINT_COMPLETE`), updates the operator's Beta posterior, and writes a bidirectional event-rating link, all within one Fabric read-write set. Fabric's commit semantics make these effects atomic, so no observable state holds the physical evidence on-chain without its reputation consequence, or the reverse. The bridge required no chaincode changes, since the existing `offChainDataHash` field absorbs the attestation layer.

V. IMPLEMENTATION

Hardware. A Creality Ender-3 FDM printer (8-bit Marlin-class controller, no secure element) is driven over USB serial by OctoPrint [15] on a Raspberry Pi.

Recorder (`capture_poll.py`, Python 3). During a job the recorder polls OctoPrint's read-only REST endpoints (`/api/printer`, `/api/job`) every 5 s and appends each sample (nozzle and bed actuals and targets, job state) to a durable JSONL log, so a crash loses at most one interval. When an observed job leaves the printing states, the recorder closes the capture and classifies the outcome as `PrintDone` at $\geq$99.5% completion or `PrintCancelled` otherwise.

Attestation module (`attestation.py`) parses the G-code envelope, assembles the canonical bundle, and manages the device key. The Ed25519 key [16] is generated on first use and persists on the bridge host as a PKCS#8 PEM file. The device identity `DEV:`h_{16} is the truncated SHA-256 of the raw public key and serves as the T2 registry key.

Ledger. Hyperledger Fabric v3.1.0 (two organizations, Raft ordering, TLS) runs the three-contract chaincode as Chaincode-as-a-Service. The submission script drives the lifecycle: `CreateMaterialCertification`

binds the asset to the certified design hash, and `RecordProvenanceWithReputation` then commits the print-job event with the bundle hash, the derived rating (Eq. (2)), and the payload's evidence hash.

VI. Evaluation

We evaluate four questions: (Q1) does the pipeline produce a verifiable, on-chain-anchored evidence record from a real print job; (Q2) does physical adherence propagate into a reputation update; (Q3) is tampering with stored evidence detected; and (Q4) what are the bridge's latency and storage costs. Each question is answered with measured data from five real print jobs on the Ender-3 testbed, spanning three geometries and three outcome types; Table II summarizes the runs.

A. Live Capture Case Study (Q1)

A print job for asset `PART-f4u2-0001` was executed on the Ender-3 from `f4u2.gcode` (4,998,971 B, SHA-256 `195eabe4...`, Cura 5.12.0, commanded envelope 200 °C nozzle / 50 °C bed). The recorder captured 115 samples at a ∼5 s interval over a 578 s window; the operator cancelled shortly after both heaters reached setpoint, giving a `PrintCancelled` exemplar.

Assessment. Applying the policy of Section IV-B: `compliance` = 1.0 (executed G-code hash matches the certified design), `delivery` = 1.0 (578 s ≤ 13,639 s estimate), and `quality` = 0.2 (46 of 230 nozzle and bed readings within ±8 °C of target), yielding $r = 0.6(0.2) + 0.25(1.0) + 0.15(1.0) = 0.52$. The low quality score reflects the idle and heat-up samples preceding setpoint, which the current policy counts against the target; steady-state tracking in this run was tight.

Commitment. The signed bundle canonicalizes to 13,408 B and its commitment (`offChainDataHash` = `9b739c98...`) is stored in the `PRINT_JOB` event, with the bundle held off-chain under device signature `DEV:1cf20afaacb6ed6b`. All five runs reproduce this property (Table II).

B. Controlled Thermal-Anomaly Comparison

This experiment compares two prints of the same calibration cube, captured from the same idle start so that heat-up samples enter both assessments identically. In `cube_anomaly`, a mid-print setpoint override (`M104 S215`, +15 °C) was injected through the OctoPrint terminal after the nozzle had stabilized at 200 °C, held for 60 s, then reverted. Figure 2 shows the excursion, with 14 samples leaving the ±8 °C tolerance band.

Against the normal print (`cube_done`, `quality` = 0.780, 103/132 in-tolerance), the anomaly run scores `quality` = 0.708 (92/130 in-tolerance), a 9.2% relative drop attributable to the injected deviation alone. `compliance` remains 1.0 across both runs, since the certified design was never altered, so the bridge assigns the degradation to physical execution rather than design substitution, a separation recovered from telemetry alone on unmodified consumer hardware.

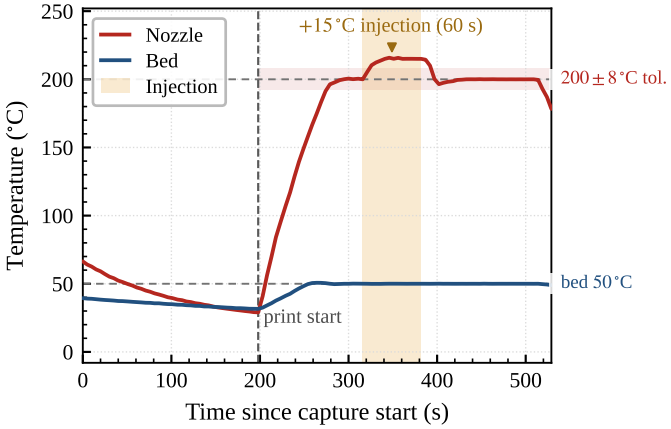


Fig. 2. Thermal telemetry from the `cube_anomaly` run (105 samples, 5 s interval). The `M104 S215` injection (shaded) drives the nozzle out of the ±8 °C tolerance band; the bed holds 50 °C.

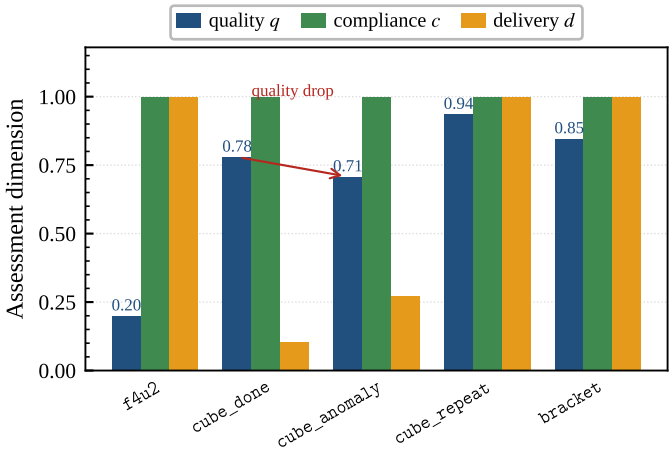


Fig. 3. Assessment dimensions across the five runs. `compliance` holds at 1.0 (the certified design was never substituted) while `quality` tracks physical execution; the `cube_done` versus `cube_anomaly` pair (identical idle-start prints) shows the injection's quality drop.

A third completed print, `cube_repeat`, scores higher (`quality` = 0.936) because its recorder attached after warm-up (∼2.6% progress), omitting the out-of-tolerance heat-up samples; it is a completed-print data point, not part of the controlled comparison.

C. Assessment Signals and Reputation Propagation (Q2)

Figure 3 plots the three assessment dimensions across the five runs. Each run's `quality` value was committed against operator `operator-secnet` through the atomic `RecordProvenanceWithReputation` call, updating an on-chain Beta posterior from the prior $\mathrm{Beta}(2.0, 2.0)$. Over the five events the posterior mean rose from 0.539 to 0.712 and the 95% Wilson interval narrowed from width 0.684 to 0.561, so attested adherence accumulates into a per-actor trust signal and sparse history is penalized, closing the selective-reporting loophole of Section III-B1. The reputation layer itself is prior work [4]; the bridge produces a signal it consumes directly.

TABLE II

THE FIVE REAL PRINT JOBS: THREE GEOMETRIES, THREE OUTCOME TYPES. ALL BUNDLES VERIFY (ED25519 SIGNATURE AND LEDGER-HASH CHECK) AND ALL CARRY `COMPLIANCE` $= 1.0$ (EXECUTED G-CODE HASH EQUALS THE CERTIFIED DESIGN). `Q`/`C`/`D` ARE THE ASSESSMENT DIMENSIONS OF EQ. (2); CANONICAL SIZE IS THE OFF-CHAIN SIGNED BUNDLE, WHOSE ON-CHAIN FOOTPRINT IS A CONSTANT 369 B REGARDLESS OF BUNDLE SIZE.

Run	Geometry	Outcome	q	c	d	rating r	samples	canon. B	on-chain B
f4u2	complex (5.0 MB)	PrintCancelled	0.200	1.0	1.000	0.520	115	13,408	369
cube_done	calibration cube	PrintDone	0.780	1.0	0.105	0.734	115	13,528	369
cube_anomaly	calib. cube (+15 °C)	PrintDone	0.708	1.0	0.271	0.715	105	12,459	369
cube_repeat	calibration cube	PrintDone	0.936	1.0	1.000	0.962	56	7,214	369
bracket	L-bracket (functional)	PrintDone	0.847	1.0	1.000	0.908	84	10,213	369

TABLE III

EMPIRICAL T1 RESULTS ON THE FABRIC AND ENDER-3 TESTBED. EACH ATTACK TARGETS A DEFENSE CLAIMED IN TABLE I.

Attack	Mechanism	Result
Cross-asset replay	Lifecycle state machine	Rejected: requires `MATERIAL_CERTIFIED` stage
Cross-design evidence	Design-hash binding	Binding mismatch; c=0, rating vetoed
Wrong-key attestation	Ed25519 verify	Wrong key fails, device key passes
In-flight modification	Hash commit and signature	Rejected twice: hash mismatch and bad signature
Design substitution (live)	Compliance veto and Beta update	Live print; c=0, r=0.50 (0.75 if compliant)

D. Adversarial Evaluation (Q3)

We exercised each T1 defense of Table I against the live Fabric and Ender-3 testbed; Table III reports the outcomes. For *post-hoc tampering*, a single stored telemetry sample was altered to 235 °C: verification failed on both independent checks, the recomputed bundle hash no longer matched the on-chain `offChainDataHash` (Eq. (1)) and the Ed25519 signature did not verify. The two checks fail under different compromises, so a compromised ledger view still fails the signature and a leaked key re-signing altered evidence still fails the hash, because the commitment covers the original signature. For *wrong-key attestation*, a bundle verified under an attacker keypair failed and passed only under the device key. For *in-flight modification*, a byte flipped between attestation and gateway fired both detections. For *cross-asset replay*, resubmitting a valid bundle hash against an asset outside the `MATERIAL_CERTIFIED` stage, and against a nonexistent asset, was rejected by the chaincode lifecycle guard.

For *design substitution* we printed a modified cube on the Ender-3 (layer height altered, executed hash $\neq$ certified hash) and committed its bundle from 100 live telemetry samples. The executed-versus-certified mismatch drove `compliance` $= 0$, which vetoes the $0.25c$ term of Eq. (2) and lowered the rating to $r = 0.50$, against 0.75 for a compliant print. The vetoed rating propagated on-chain in the same atomic transaction, reducing the operator's positive posterior contribution for that event by roughly one third.

Two enforcement boundaries in the current build are worth stating precisely. The lifecycle stage guard and the design-hash binding are applied on the `RecordPrintJob` and client paths respectively; migrating both into chaincode endorsement is the T2 step. Evading the T1 checks together still requires re-signing with the device key *and* rewriting committed Fabric state against the endorsement policy, which is the fabrication-at-source residual T1 does not claim to cover.

E. Latency and Storage Cost (Q4)

Bridge latency (Q4a). We instrumented each pipeline stage over 100 iterations on each of the four PLA-profile runs (cube $\times 3$, bracket). The attest pipeline (parse, payload assembly, canonicalization, Ed25519 signing, and hash derivation) completes in a 0.91 ms mean, and adding independent verification brings the round trip to 1.55 ms, with signing (0.32 ms) and verification (0.63 ms) the dominant stages for Ed25519 over a $\sim$13 kB payload. This is small beside the ledger commit, which we measured at 227.8 ms mean (n=20 atomic provenance-and-reputation writes) on our testbed, so attestation adds under one percent to end-to-end latency. Parsing is the one size-dependent stage: 0.08 ms for the $\sim$20–38 kB cube and bracket files but 11.4 ms for the 5 MB `f4u2.gcode`, because parsing scales with design size while signing operates on the fixed-size hash.

Bundle-size scaling (Q4b). Canonical bundle size grows linearly with retained samples, from 7.2 kB (56 samples) to 13.5 kB (115 samples) across the five runs, while the on-chain footprint stays at 369 B regardless of bundle size (Figure 4), a 19.5$\times$ to 36.7$\times$ reduction from the off-chain bundle to its on-chain commitment. Extrapolating linearly, multi-hour prints at 5 s sampling reach a few MB off-chain while the ledger cost stays fixed (Section VII).

VII. DISCUSSION

A. Guarantees and Boundary of T1

The captured runs support the T1 claim: once the device signs, every byte of evidence is bound to the ledger, and single-sample tampering is caught by two independent checks. They also mark its boundary. Had the bridge host synthesized the entire trace before signing, nothing downstream would reveal it. A certifying authority should therefore read a T1 record as evidence that has not changed since the device signed it, not as

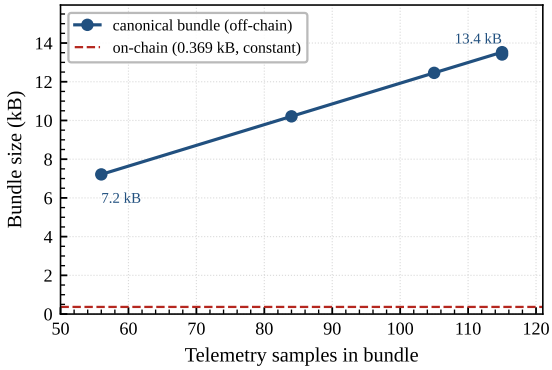


Fig. 4. Off-chain bundle size grows linearly with retained samples (five runs), while the on-chain footprint stays constant at 369 B.

proof that the print occurred as described. For the threats that dominate in practice (Section III-B1), tamper-evidence after signing is the property required.

B. Path to T2 and T3

T2 requires no new cryptography. The device identity and raw public key are already in the bundle, and the chaincode adds a registry keyed by device ID with enrollment, revocation, and endorsement-time Ed25519 verification. Enrollment becomes a ledger event under the endorsement policy, so a stolen key is bounded by its revocation window and a bureau cannot quietly add signing devices. T3 targets fabrication at source by moving the key into a TEE or secure element, with measured boot of the recorder and, on industrial machines, signing at the sensor or controller level; its strongest form depends on printer-side hardware the consumer fleet does not have.

C. Limitations

Even T3 with sensor-level signing attests only that a reading came from a given device, not that the sensor measures the quantity it claims to. A detached thermistor defeats any downstream cryptography, and no tier controls where a sensor is placed, so attestation bounds the cost of fabrication without making the physical process self-authenticating. The empirical scope is a second limit: five jobs on one printer across three geometries and three outcomes, with breadth across machines and sites as the next step. Two prototype choices remain within T1. The recorder polls OctoPrint's REST interface rather than its event bus, which keeps the bridge printer-agnostic but coarsens outcome classification, and the 5 s poll interval, sufficient for the injected anomaly, can be shortened for finer transients; an in-process event-hook plugin resolves both. Off-chain bundle size grows with print duration while the on-chain footprint stays fixed; windowed summarization is a future extension of the bundle format.

VIII. Conclusion

On-chain provenance for additive manufacturing is only as trustworthy as the bridge that carries physical evidence onto the ledger. We presented a four-tier trust analysis of that bridge, a canonical device-signed telemetry bundle anchored by a single on-chain hash, and a working T1 implementation on unmodified consumer hardware integrated with a Fabric v3.1.0 provenance and reputation chaincode. Across five real print jobs, a controlled $+15\,^{\circ}\mathrm{C}$ injection lowered the quality score by 9.2% while compliance held at 1.0, and all five adversarial tests were detected or rejected at a 1.55 ms round-trip cost. The remaining gap, that T1 cannot detect fabrication at source by the device-key holder, sets the agenda for future work: a T2 on-chain device registry, T3 hardware-rooted or threshold attestation, and evaluation across machines and sites.

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